Learned representations

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Image-pattern Network Investigation

THE CONNECTED WEEK

Make the available information and target conflict visible.

Use complete pairs,areas and encodings;separate mathematical sufficiency from claims that training learned a concept.

Keep one habit: Assess independent scores,shared hidden vector and conflicting targets with checked keys.

MAIN LESSON · SECTION 1

Prepare the connected mathematical investigation

> Keep explanation,shared practice and independent evidence distinct.

Target and main teaching plan

Primary target: Trace an **input** through a hidden **representation** and a decision rule

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min

> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Words that unlock the idea and Follow a worked example; record recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Use Find and repair a mistake, Practise together and Build your understanding; use the matching workbook W2 and Individual reasoning check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project and Try a new case independently. Allow extra time for the full task set. When code is supplied, use the examples in See how it works or Follow a worked example and the browser lab in Supported practice, repair and portfolio.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Preparation and prerequisite

Printed lesson and workbook; paper and pencil

Calculator and optional local Python 3 editor; no packages or AI accounts needed

Use the fictional examples supplied; prepare the fresh retry before class

Different negative scores all become zero. Matching activations can hide an earlier error, so intermediate values need independent checks.

Opening retrieval

Without opening the old lesson, explain one key idea from Debugging shapes and gradients. Give an example and a mistake that the idea helps you avoid.

Look for: Check dimension contracts before computing a layer

Reference example, not the only acceptable wording: Teacher explanation: No. Each weight row must have the same length as x.

Confirm the intended feature schema, then supply a valid two-feature **input** or deliberately redesign and retrain the layer; do not silently discard a value.

Without opening the old lesson, explain one key idea from Regularisation and early stopping. Give an example and a mistake that the idea helps you avoid.

Look for: Separate a weight penalty from **prediction** error

Reference example, not the only acceptable wording: Teacher explanation: MSE 0, penalty 2, total J 2, total gradient 2. Zero data gradient does not remove the penalty gradient.

Without opening the old lesson, explain one key idea from Batching and epochs. Give an example and a mistake that the idea helps you avoid.

Look for: Distinguish an example a batch an update and an epoch

Reference example, not the only acceptable wording: Teacher explanation: Batch lengths 4, 4, 2; three updates per epoch, nine total updates and 30 example visits.

Pictured end goal

By the end, I can explain learned **representation**s using a traced example and a stated limitation.

▼ Earlier wording, alternatives and picture transcripts

Teach the learning cycle

Retrieval answer guidance

Debugging shapes and gradients

Regularisation and early stopping

Batching and epochs

For the compact alternative schedule, use two 45-minute sessions. Begin with rectangle measurements and the explicit strict comparison. Distinguish the constructed layer from a genuinely fitted layer throughout.

The key reasoning is the **collision** argument: identical **representation**s force identical deterministic **output**s. No calculus, external AI account or full **training** framework is required.

Outcomes

Trace an **input** through a hidden **representation** and a decision rule

Identify distinctions that a **representation** preserves or loses

Separate a useful constructed feature from **evidence** of successful learning

Preparation

Prerequisite answer

End goal for the pictorial case

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

Teach the target distinction before judging a representation

> Main rectangle and pictured brightness encoders share a structure but name different measurements.

Main formulas,complete trace and guided key

Consider fictional rectangles with **input** [width, height] in cm. The task is to identify rectangles whose width is strictly greater than height. The target is 1 for wider rectangles and 0 otherwise.

A square therefore has target 0. This is an arithmetic teaching task, not a claim that a network is needed to measure rectangles.

Define two hidden units: h1 = ReLU(width - height) and h2 = ReLU(height - width). Their weight rows are [1, -1] and [-1, 1], with zero biases. The **representation** h = [h1, h2] records excess width or excess height.

Predict 1 when h1 > 0; otherwise predict 0.

These weights were chosen by the lesson author. They demonstrate a **representation** that could be implemented by a network, but no **training** run has established that a model learned it.

Learning would require adjusting parameters using **training** data and recording the procedure.

A **prediction** head is the computation that reads the hidden vector and produces an **output**. Keeping it separate lets us inspect what information is available before making a decision.

This example assumes two finite, positive measurements in cm; it is not a general **input** validator.

The raw difference width - height can be recovered as h1 - h2. The original width and height cannot in general be recovered: many rectangles share a difference.

Two hidden components do not guarantee preservation of two **input** components.

Input[width, height]in cm · Scores[width–height, height–width] · Hidden hafterReLU · Wider prediction · Area in cm²

> **Input[width, height]in cm:** [4, 2]

> **Scores[width−height, height−width]:** [2, −2]
>
> **Hidden hafterReLU:** [2, 0]
>
> **Wider prediction:** 1
>
> **Area in cm²:** 8

> **Input[width, height]in cm:** [2, 4]
>
> **Scores[width−height, height−width]:** [−2, 2]
>
> **Hidden hafterReLU:** [0, 2]
>
> **Wider prediction:** 0
>
> **Area in cm²:** 8

> **Input[width, height]in cm:** [3, 3]
>
> **Scores[width−height, height−width]:** [0, 0]
>
> **Hidden hafterReLU:** [0, 0]
>
> **Wider prediction:** 0
>
> **Area in cm²:** 9

> **Input[width, height]in cm:** [5, 3]
>
> **Scores[width−height, height−width]:** [2, −2]
>
> **Hidden hafterReLU:** [2, 0]
>
> **Wider prediction:** 1
>
> **Area in cm²:** 15

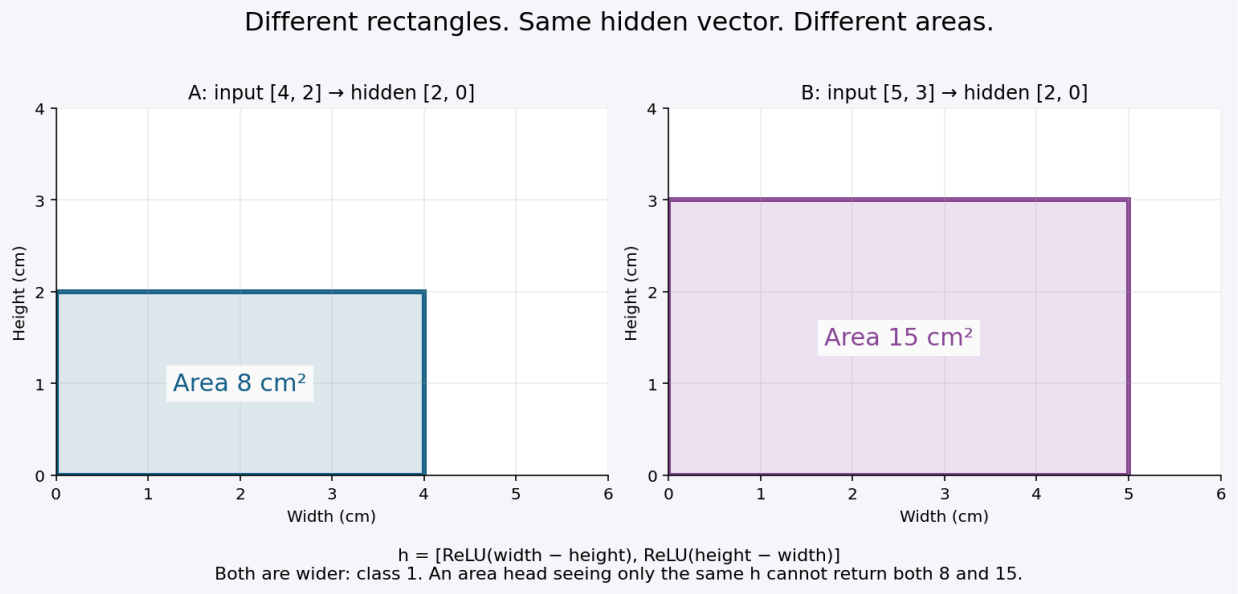
A collision means different original inputs share the same h. Whether that matters depends on the target. Rectangles[4, 2]and[5, 3]both should receive wider-label 1, so their collision does not obstruct that comparison. Their required areas 8 and15 differ, so a deterministic area head receiving only h cannot return both exactly. Keeping two hidden components did not preserve the original two measurements.

This is an algebraic property of the supplied encoder, not a measured learned feature or generalization result.

Guided: [6, 4] and [7, 5] both encode as [2, 0], predict 1 and have areas 24 and 35 square cm. A deterministic head using only h cannot recover both exact areas.

Exit: hand-chosen weights are not **evidence** of learning, and correct comparison labels do not establish area **prediction**. Test the **collision** [4, 2] versus [5, 3], with areas 8 and 15.

Scale-matched collision and complete information repair



Complete rectangle repair	Information / result
Original h=[excess width, excess height]	Recovers difference d=h1−h2, but not size
Only sum s=width+height	[4, 2]and[2, 4]both s6 but wider targets 1/0; sum alone fails that task
Retain hplus s	width=(s+d)/2, height=(s−d)/2
h[2, 0], s6	d2, width 4, height 2, area 8
Sameh[2, 0], s8	d2, width 5, height 3, area 15

Complete rectangle repair	Information / result
Scope	Exact valid compatible measurements; algebraic recovery is not proof of successful training or noisy-data generalization

Separate brightness facts and scope

See when a **representation** preserves a distinction and when it destroys it.

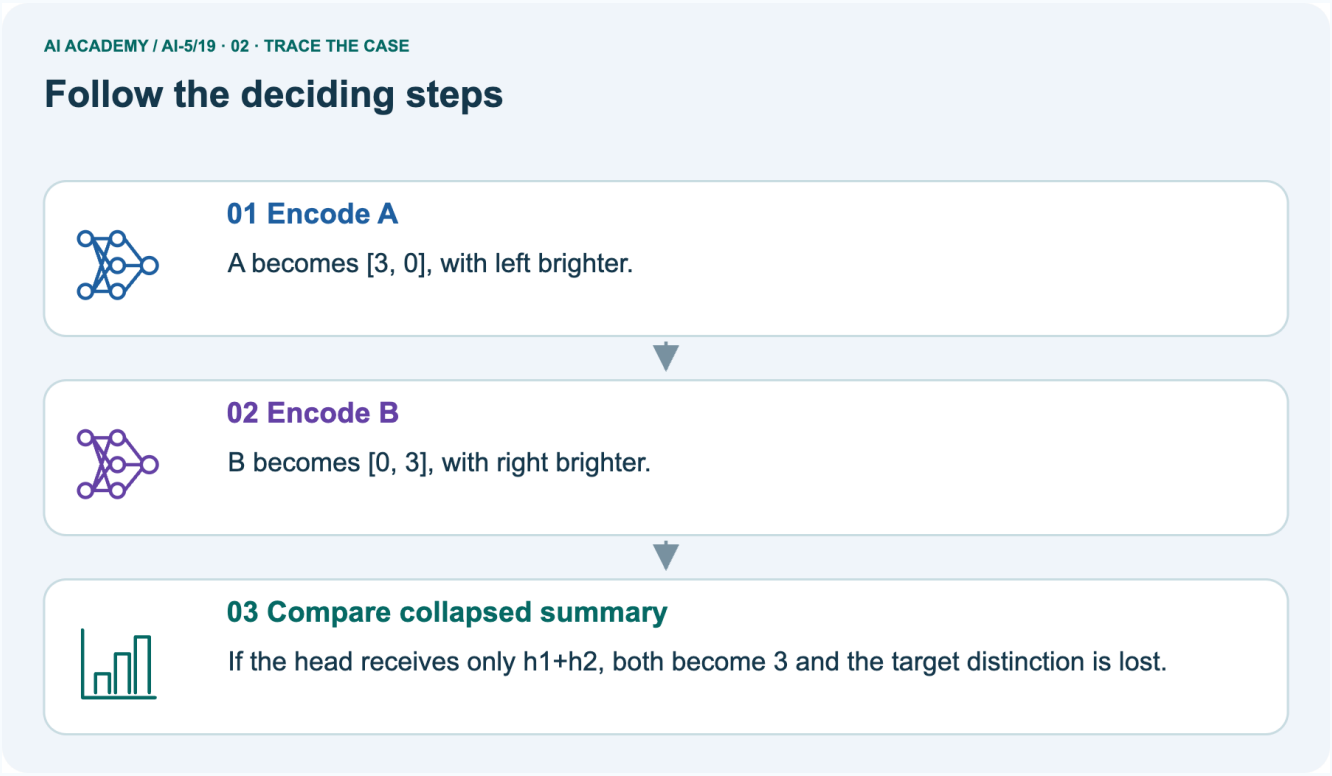
Toy **input** [left, right]; **representation** h=[ReLU(left-right), ReLU(right-left)]. A=[5, 2]; B=[2, 5]; target is left-brighter versus right-brighter.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

The picture uses left/right brightness, not rectangle dimensions. A[5, 2]encodes[3, 0], B[2, 5]encodes[0, 3]. Their ordered vectors preserve which side is brighter. Passing only h1+h2 gives3 for both and loses the target direction, so no deterministic sum-only head can assign both different labels correctly. Keeping both ordered values restores this available distinction for this pair.

These weights are hand-selected; no trained vision model is claimed.

Trace ordered information versus collapsed sum



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Encode A	A becomes [3, 0], with left brighter.
2. Encode B	B becomes [0, 3], with right brighter.
3. Compare collapsed summary	If the head receives only h1+h2, both become 3 and the target distinction is lost.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

A hidden **representation** decides which distinctions remain available to an **output** head. These two hidden coordinates preserve whether the left or right side is brighter.

Adding them discards that direction and gives both examples the same scalar. A deterministic head cannot assign different labels to identical **input**s; retain the distinction or change the task.

This is a hand-chosen **representation**, not **evidence** that a trained image model actually learned these units.

Change: The head keeps both ordered hidden values instead of their sum.

Prediction: It can distinguish [3, 0] from [0, 3] in this example.

Reason: The **representation** retains the direction of brightness difference when coordinates stay separate.

▼ Earlier wording, alternatives and picture transcripts

Term	Meaning
Representation	An encoding of an example used by another computation
Hidden activation	A value produced inside a network after applying an activation function
Collision	Two different inputs receiving the same representation
Learned feature	A feature whose computation depends on parameters adjusted using training data

Different rectangles can share a hidden vector

INPUT IN CM	HIDDEN VECTOR	AREA IN SQUARE CM
Width 4 height 2	[2, 0]	8
Width 5 height 3	[2, 0]	15

The representation preserves the wider-than-height answer for these examples but discards information needed to recover their different areas. A deterministic head receiving only this vector must give both examples the same output.

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Different rectangles can share a hidden vector

INPUT IN CM	HIDDEN VECTOR	AREA IN SQUARE CM
Width 4 height 2	[2, 0]	8
Width 5 height 3	[2, 0]	15

The representation preserves the wider-than-height answer for these examples but discards information needed to recover their different areas. A deterministic head receiving only this vector must give both examples the same output.

Encode [6, 4] and [7, 5]. Give their wider-than-height predictions and areas. Can a head that sees only h always recover the exact area? Explain using this pair.

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Learned representations

The facts we will use

AI ACADEMY / AI-5/19

A representation can preserve direction

	left	right	h1	h2
A	5	2	3	0
B	2	5	0	3

Read a cell using both its row and column labels.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-5/19

Learned representations

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Encode A

A becomes [3, 0], with left brighter.



2. Encode B

B becomes [0, 3], with right brighter.



3. Compare collapsed summary

If the head receives only h1+h2, both become 3 and the target distinction is lost.

CHANGE ONE THING

The head keeps both ordered hidden values instead of their sum.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

Change one thing in the pictured case

Ask while showing the picture

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a **representation**; it does not document execution of a real AI model.

MAIN LESSON · SECTION 3

Model a step,then remove help

> Collect each learner’s reasoning before showing a key.

Current completion task and checked reasoning

Completion task: Give the hidden vectors and areas for [4, 2] and [5, 3]. Explain why increasing only the complexity of a deterministic area head cannot make both exact **prediction**s correct.

Teacher check: Teacher explanation: Both hidden vectors are [2, 0], while areas are 8 and 15 square cm. Identical **input**s to a deterministic head yield identical **output**s, so the head cannot return both required values.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual misconception check

Use the worked case for Learned **representation**s. Proposed explanation: “Two hidden values preserve both **input** measurements”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: Dimension alone does not guarantee recovery; distinct **input**s can collide.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictured reasoning and changed case

Why can no deterministic head using only the sum classify both correctly? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Both examples supply the identical scalar 3 but require different targets. Identical **input**s cannot produce different deterministic **output**s. Changed-case check: It can distinguish [3, 0] from [0, 3] in this example.

The **representation** retains the direction of brightness difference when coordinates stay separate.

Changed-case reasoning: It can distinguish [3, 0] from [0, 3] in this example. The **representation** retains the direction of brightness difference when coordinates stay separate.

Collect each learner’s explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

Accessible support and extension

Use See how it works and Follow a worked example. Collect a **prediction** before revealing its result; ask the learner to point to each intermediate fact.

Provide paper rectangles and columns for signed difference, two scores, activations and target. Allow calculators and ask for verbal **collision** explanations before algebra.

Extension: show that adding the same positive amount to width and height preserves h while changing area; derive exact recovery when the sum is also retained.

▼ Earlier wording, alternatives and picture transcripts

Time or stage	Teacher action
Session 1 start to 7 minutes	Retrieve how ReLU can hide distinctions and establish the rectangle schema and target.
7 to 18 minutes	Trace the four lesson examples, including a square, before naming the hidden features.
18 to 32 minutes	Complete W1 and trace the optional code, separating the representation from its head.
32 to 45 minutes	Use the visual and W2 to explain why identical hidden inputs cannot yield two exact areas.
Session 2 start to 8 minutes	Complete the guided pair and compare the sum-only encoding using W3.
8 to 20 minutes	Derive recovery from sum and difference, then discuss the limited conclusion in W5.
20 to 35 minutes	Collect W4 independently, requiring both score vectors, hidden vectors and target labels.
35 to 45 minutes	Use the exit response and fresh retry to distinguish learning evidence from an authored formula.

Reduce help deliberately

Individual reasoning check

Pictorial practice question

Reasoned teacher answer

Run the complete supplied investigation

> Separate predictions,actual runs and paper traces.

Complete inputs,expected results and limits

Main collisions and exact repair as supplied. W2[4, 2]/[5, 3]samehbutareas 8/15, nohead-only complexityrepair. W3 sum6 for[4, 2]/[2, 4]withtargets 1/0. W5h[2, 0], s8→width 5, height 3, area 15, butnolearning/noisyperformanceproof. Independentg=[ReLU(w−h−1), ReLU(h−w−1)]: [4, 3]scores[0, −2], g[0, 0], target 1; [3, 4]scores[−2, 0], g[0, 0], target 0. No deterministicg-onlyheadcanclassifyboth.

Require target-dependent collision reasoning and constructed-versus-learned distinction.

Diagnose the current misconceptions

Two hidden values preserve both **input** measurements: Dimension alone does not guarantee recovery; distinct **input**s can collide.

A more complex head can always fix lost information: A deterministic head cannot distinguish identical **representation**s without additional **input**.

Correct **output**s prove human concepts were learned: The layer here was constructed, and a task score alone does not establish a hidden concept.

Use Run the investigation for the **evidence** ledger. Add a boundary or missing-**input** case and record the expected response before checking. More successful repeats of one case do not establish wider coverage.

Choose accessible support

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.

What the response shows	Next teaching action
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

▼ **Earlier wording, alternatives and picture transcripts**

Match the support to the first error

MAIN LESSON · SECTION 5

Assess the new schema or calculation transfer

> Keep the selected answer private until after a first attempt.

Selected independent task and checked answer

Selected case: A different layer uses g = [ReLU(width - height - 1), ReLU(height - width - 1)]. Encode [4, 3] and [3, 4]. State their targets for width strictly greater than height.

Can any deterministic head receiving only g classify both correctly?

Teacher explanation: Both encode as [0, 0]: the scores are [0, -2] and [-2, 0]. Targets are 1 and 0. No deterministic head receiving only this shared vector can distinguish the pair.

The added threshold erased a distinction required by this task.

Judge explanation,trace and limits

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Give one usable repair

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

W1 Trace new inputs

Encode [8, 3], [3, 8] and [4, 4] with the lesson layer. Apply its **prediction** rule and explain the square boundary.

Teacher explanation: The vectors are [5, 0], [0, 5] and [0, 0]. **Prediction**s are 1, 0 and 0. Equal width and height does not satisfy strictly greater.

W2 Prove an information loss

Give the hidden vectors and areas for [4, 2] and [5, 3]. Explain why increasing only the complexity of a deterministic area head cannot make both exact **prediction**s correct.

Teacher explanation: Both hidden vectors are [2, 0], while areas are 8 and 15 square cm. Identical **input**s to a deterministic head yield identical **output**s, so the head cannot return both required values.

W3 Compare another encoding

The alternative **representation** is s = width + height. Use [4, 2] and [2, 4] to decide whether s alone always supports the wider-than-height task.

Teacher explanation: Both have s = 6, but their targets are 1 and 0. This **collision** prevents any deterministic head using only s from getting both right.

W5 Repair and qualify a claim

For h = [2, 0] and s = 8, recover width, height and area. Does this algebra prove that a **training** program learned the **representation** or that it will work well on noisy measurements?

Teacher explanation: Difference d is 2. Width is (8 + 2) / 2 = 5 and height is (8 - 2) / 2 = 3; area is 15 square cm.

The algebra establishes recovery under the formula, not a **training** history or performance with noisy measurements.

▼ Earlier wording, alternatives and picture transcripts

Guided and exit questions

Guided: [6, 4] and [7, 5] both encode as [2, 0], predict 1 and have areas 24 and 35 square cm. A deterministic head using only h cannot recover both exact areas.

Exit: hand-chosen weights are not **evidence** of learning, and correct comparison labels do not establish area **prediction**. Test the **collision** [4, 2] versus [5, 3], with areas 8 and 15.

W1 Trace new inputs

W2 Prove an information loss

W3 Compare another encoding

W4 Audit a new representation independently

A different layer uses g = [ReLU(width - height - 1), ReLU(height - width - 1)]. Encode [4, 3] and [3, 4]. State their targets for width strictly greater than height.

Can any deterministic head receiving only g classify both correctly?

Teacher explanation: Both encode as [0, 0]: the scores are [0, -2] and [-2, 0]. Targets are 1 and 0. No deterministic head receiving only this shared vector can distinguish the pair.

The added threshold erased a distinction required by this task.

W5 Repair and qualify a claim

Independent assessment and fresh retry

W4 requires scores [0, -2] and [-2, 0], hidden vector [0, 0] for both, targets 1 and 0 and an impossibility explanation. Fresh retry: use g = [ReLU(width - height - 2), ReLU(height - width - 2)] for [5, 3] and [3, 5].

Scores are [0, -4] and [-4, 0]; both hidden vectors are [0, 0], but targets are 1 and 0. No deterministic head using only g can classify both correctly.

Assess independent transfer

Selected case answer

Feedback that the learner uses

MAIN LESSON · SECTION 6

Connect,support and revisit the mathematics

> Use the saved evidence to choose the next step.

Project and next-week bridge

Add a labelled learned **representation**s **evidence** sheet to Image-pattern Network Investigation. Show how you can trace an **input** through a hidden **representation** and a decision rule.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Provide paper rectangles and columns for signed difference, two scores, activations and target. Allow calculators and ask for verbal **collision** explanations before algebra.

Extension: show that adding the same positive amount to width and height preserves h while changing area; derive exact recovery when the sum is also retained.

Week 20 introduces image tensors and channels, with explicit axis order and value scales.

Retain the project ledger and assess a changed case after support. Check the relevant prerequisite before carrying this component forward.

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Delayed return and support record

Suggested return points in this level: ai-5/20, ai-5/22, ai-5/25. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	

Record	Teacher evidence
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

▼ Earlier wording, alternatives and picture transcripts

Plan the delayed return

Project evidence from this case

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Alternative planning checklist**

REFERENCE · SECTION 7

Reference · Alternative planning checklist

> Use alongside the main route when helpful.

Planning,assessment and project reminders

Teaching target: Trace an **input** through a hidden **representation** and a decision rule

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. Check a local Python 3 environment before class; have a paper-trace version ready.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Words that unlock the idea and Follow a worked example; record recall in the workbook.

> **Session:** Session 2 · Practise · 45 min

> **Plan:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
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> **Resources:** Use Find and repair a mistake, Practise together and Build your understanding; use the matching workbook W2 and Individual reasoning check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Plan:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project and Try a new case independently. Allow extra time for the full task set. When code is supplied, use the examples in See how it works or Follow a worked example and the browser lab in Supported practice, repair and portfolio.

Before class, solve the supported practice: Give the hidden vectors and areas for [4, 2] and [5, 3]. Explain why increasing only the complexity of a deterministic area head cannot make both exact **prediction**s correct.

Reasoned practice answer: Teacher explanation: Both hidden vectors are [2, 0], while areas are 8 and 15 square cm.

Identical **input**s to a deterministic head yield identical **output**s, so the head cannot return both required values.

Check every learner before discussion: Use the worked case for Learned **representation**s. Proposed explanation: “Two hidden values preserve both **input** measurements”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Misconception repair: Dimension alone does not guarantee recovery; distinct **input**s can collide.

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extension: The head keeps both ordered hidden values instead of their sum. Require a **prediction** and a reason before comparing. Expected result: It can distinguish [3, 0] from [0, 3] in this example.

Reason: The **representation** retains the direction of brightness difference when coordinates stay separate.

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint: Evaluate and revise. Compare with your starting method, explain errors and record a justified revision.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

▼ **Reference · Feedback,labs and repair paths**

REFERENCE · SECTION 8

Reference · Feedback,labs and repair paths

> Retain the first attempt and the evidence used for a repair.

Feedback and record guidance

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

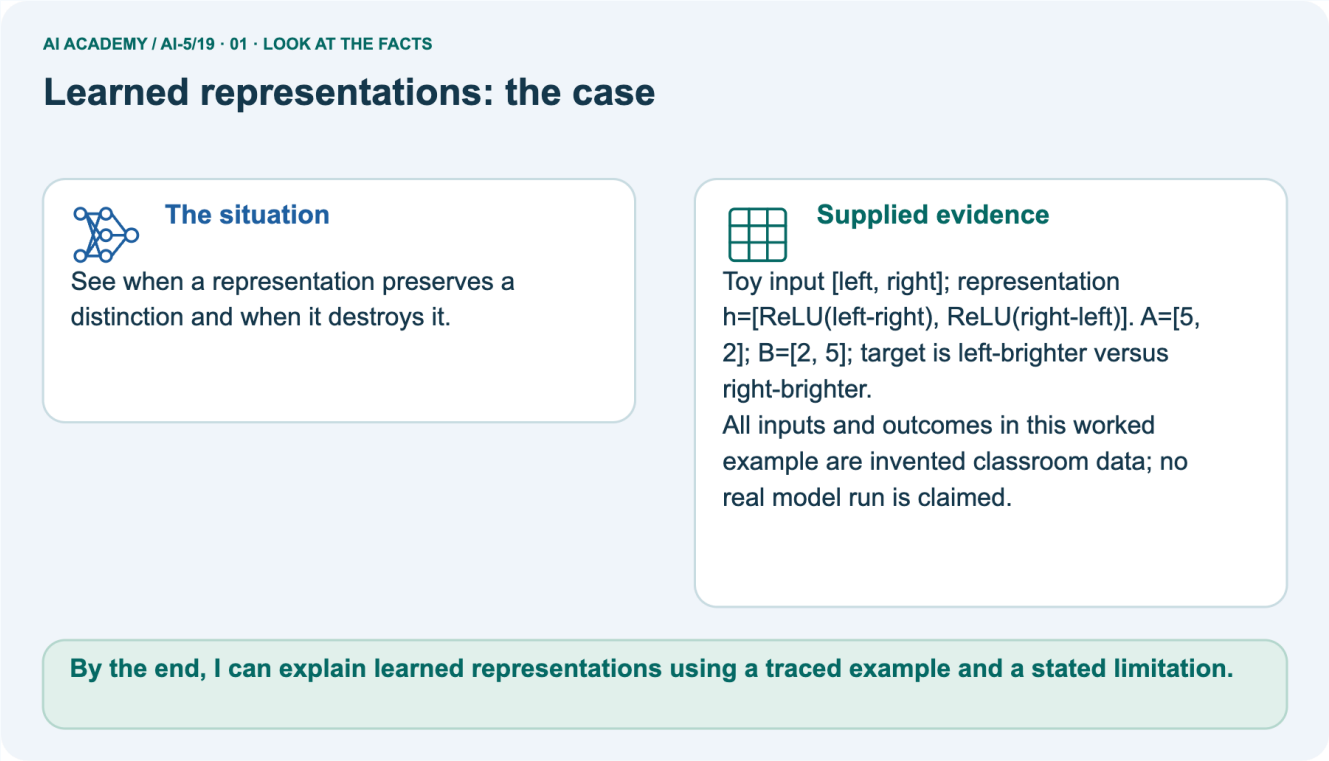
▼ **Reference · Teach with the six pictures**

REFERENCE · SECTION 9

Reference · Teach with the six pictures

> Choose one picture for a particular reasoning move.

01. Learned representations: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

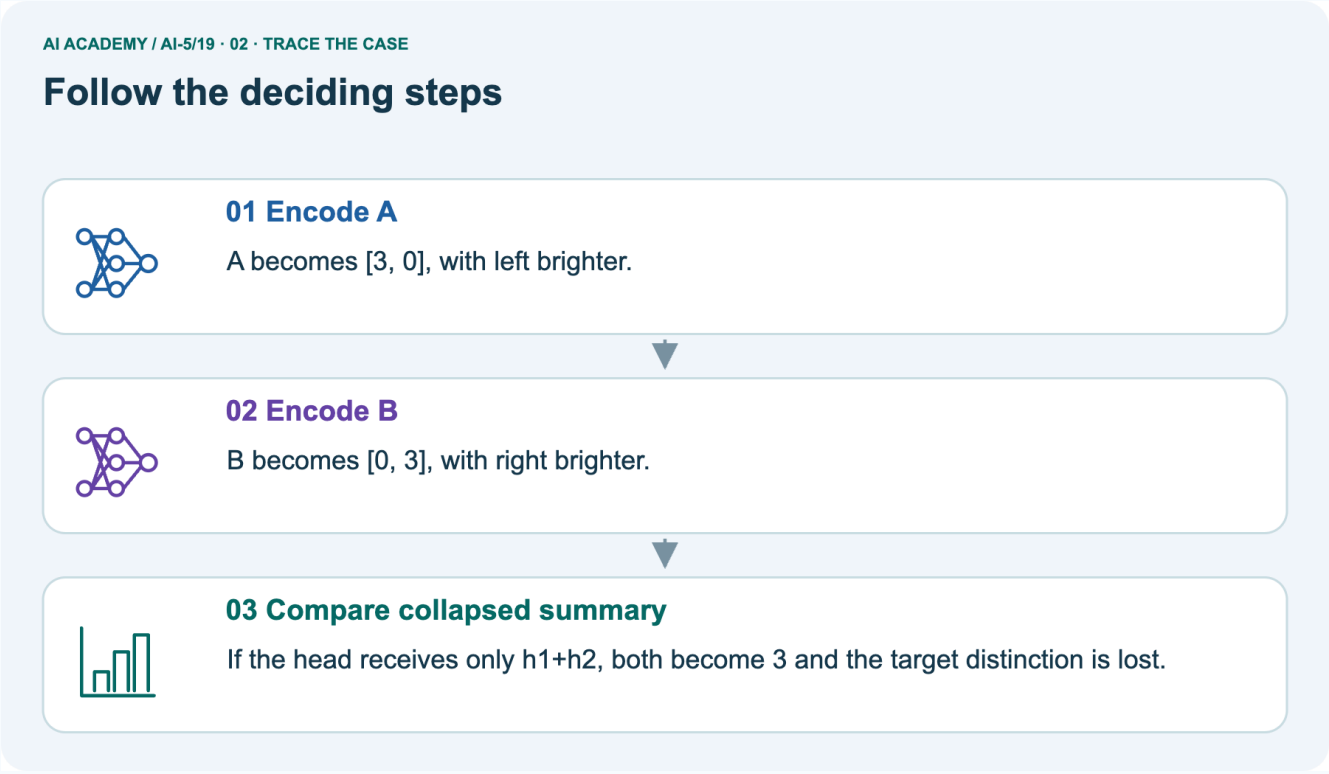
See when a **representation** preserves a distinction and when it destroys it. Supplied facts: Toy **input** [left, right]; **representation** h=[ReLU(left-right), ReLU(right-left)].

A=[5, 2]; B=[2, 5]; target is left-brighter versus right-brighter. All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



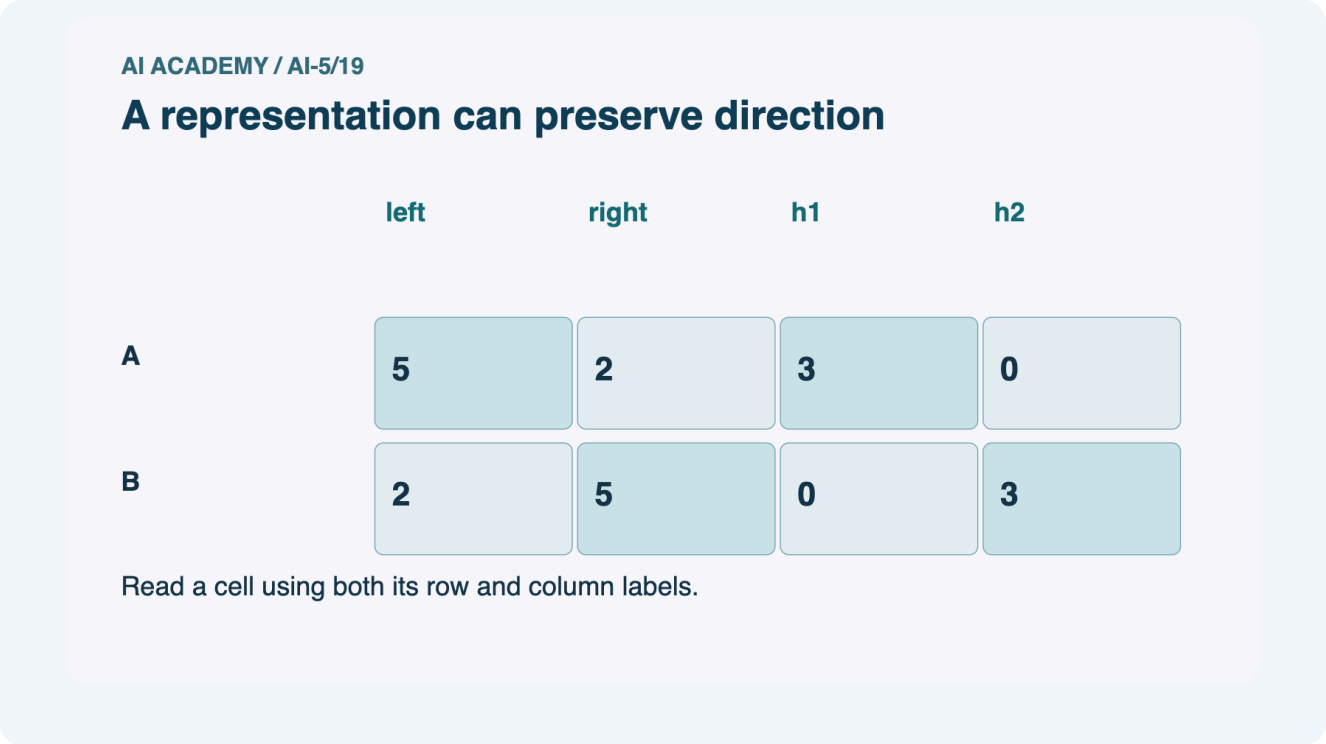
Follow the labelled picture with the matching explanation. Open the vector diagram

Encode A: A becomes [3, 0], with left brighter. → Encode B: B becomes [0, 3], with right brighter. → Compare collapsed summary: If the head receives only h1+h2, both become 3 and the target distinction is lost.

Point to each numbered stage and explain which recorded **input** it uses.

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. A representation can preserve direction



Follow the labelled picture with the matching explanation. Open the vector diagram

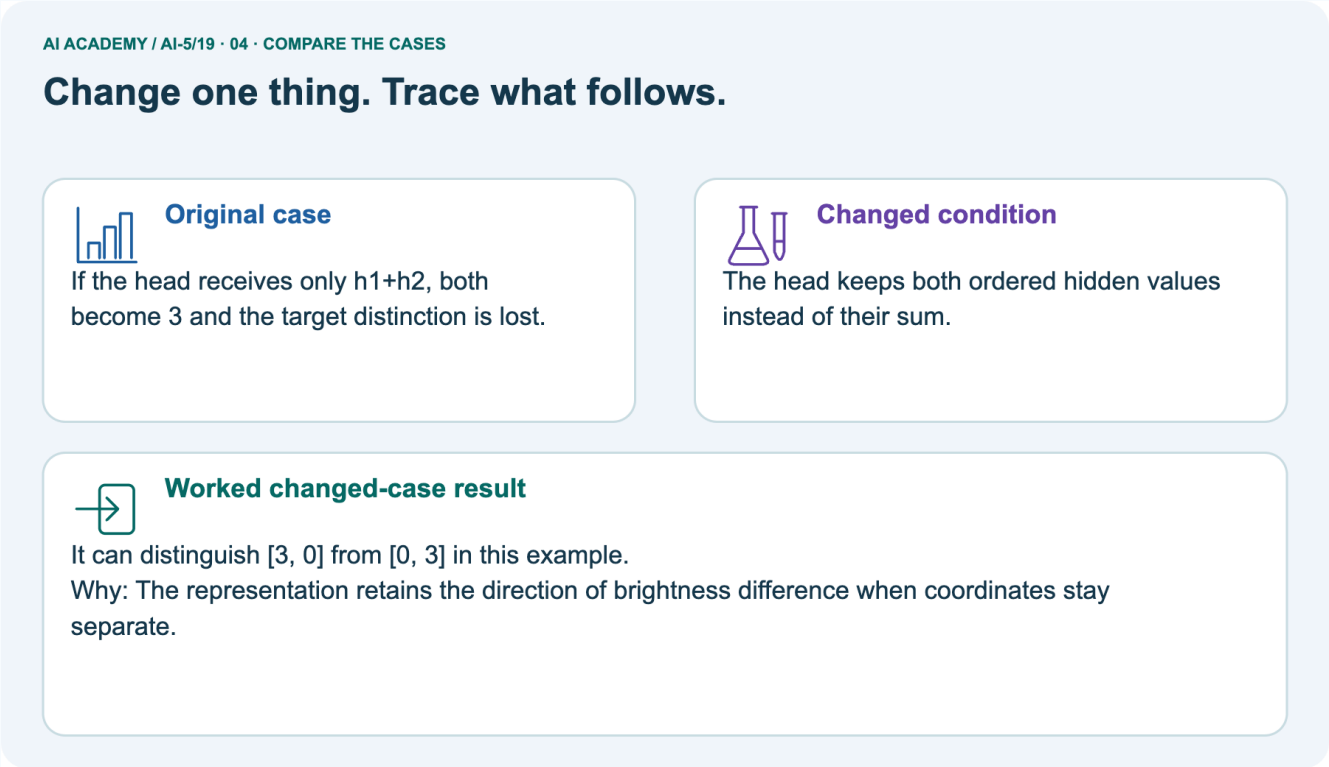
A **representation** can preserve direction. A: left = 5, right = 2, h1 = 3, h2 = 0. B: left = 2, right = 5, h1 = 0, h2 = 3.

Read the labels and values. Explain the relationship the model shows.

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Check row/column meanings, units and denominators before discussing a difference. Supplied toy values are not real class or model results.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: If the head receives only h1+h2, both become 3 and the target distinction is lost. Changed condition: The head keeps both ordered hidden values instead of their sum.

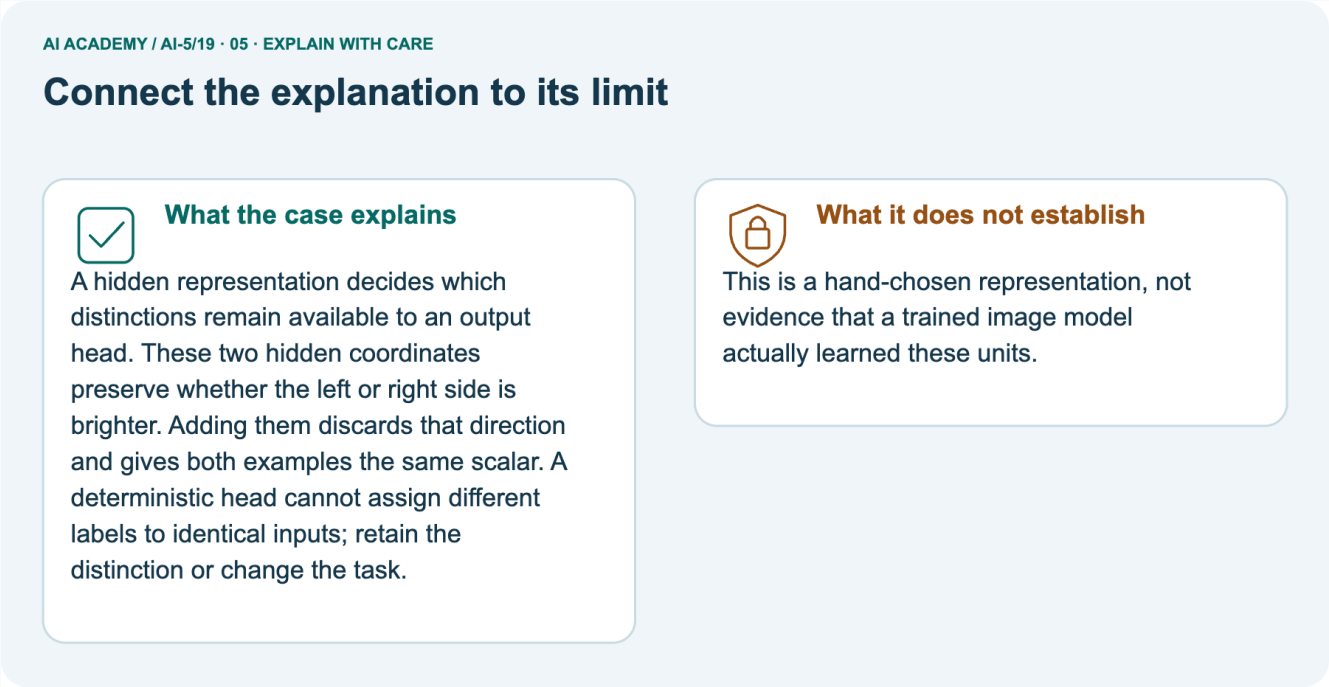
Expected result: It can distinguish [3, 0] from [0, 3] in this example. Reason: The **representation** retains the direction of brightness difference when coordinates stay separate.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: The **representation** retains the direction of brightness difference when coordinates stay separate.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

A hidden **representation** decides which distinctions remain available to an **output** head. These two hidden coordinates preserve whether the left or right side is brighter.

Adding them discards that direction and gives both examples the same scalar. A deterministic head cannot assign different labels to identical **input**s; retain the distinction or change the task.

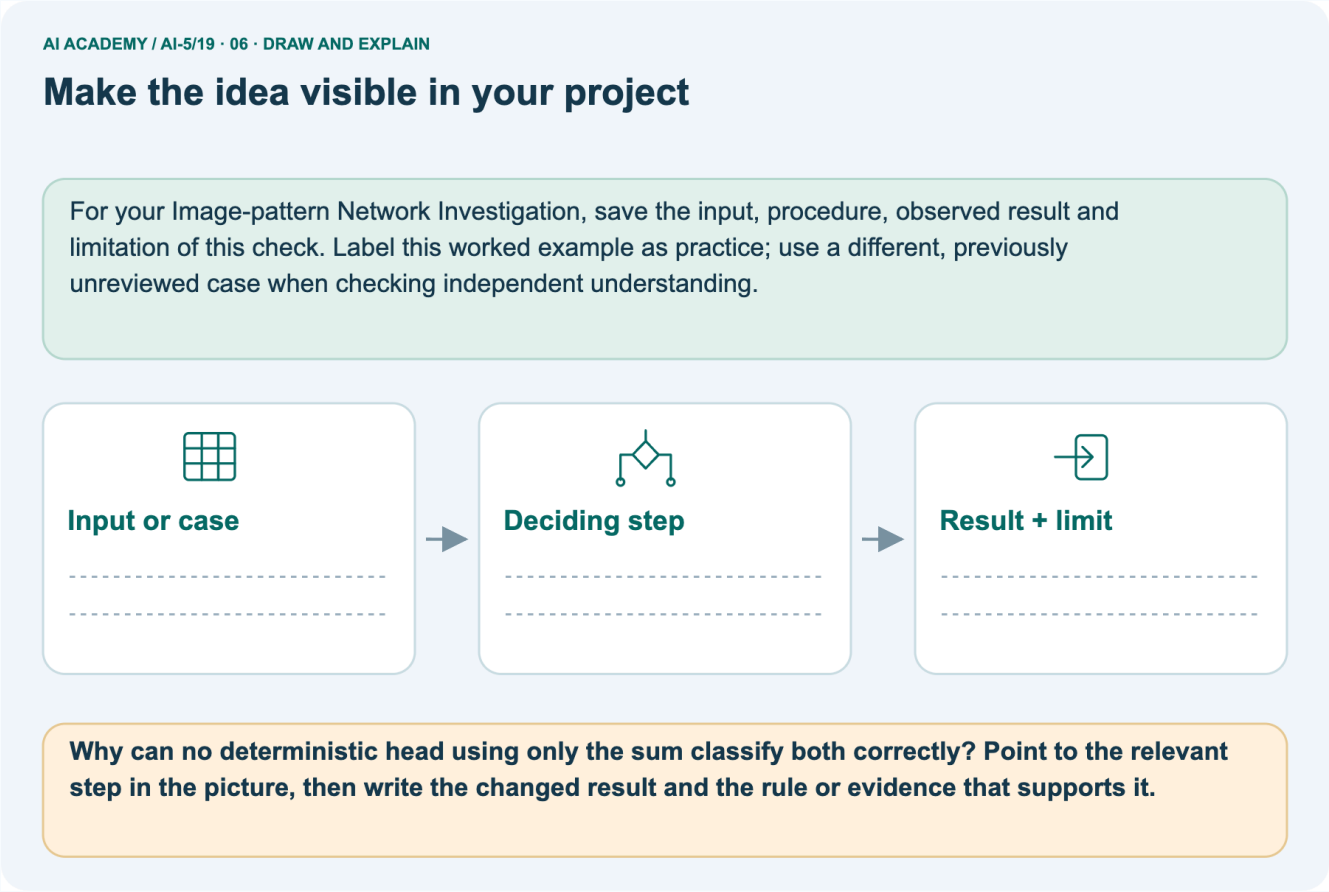
Limit: This is a hand-chosen **representation**, not **evidence** that a trained image model actually learned these units.

Explain the case, then point to a limit on the claim.

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: This is a hand-chosen **representation**, not **evidence** that a trained image model actually learned these units.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Why can no deterministic head using only the sum classify both correctly? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: Both examples supply the identical scalar 3 but require different targets. Identical **input**s cannot produce different deterministic **output**s.

Changed-case check: It can distinguish [3, 0] from [0, 3] in this example. The **representation** retains the direction of brightness difference when coordinates stay separate.

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Learned representations: the case

02. Follow the deciding steps

03. A representation can preserve direction

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

</details>

<details>
<summary>Optional interactive practice and completed work</summary>

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.
</details>